An Enhanced Virtual Robotic System with Gaze-based, EEG Brain Machine Interface for Emulated Control

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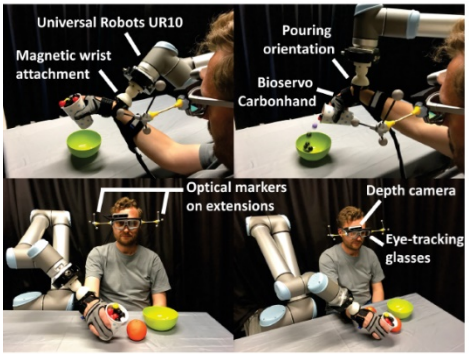
1. Abstract

Robotic control can be leveraged as a substitute for humans with motor disabilities and impairments such as paralysis, paraplegia, limb amputation or ALS to complete a physical task. This human-machine interface which alleviates the dependency on muscle motor control has been combined with various other physiology modalities as an input for robotic control. Previous studies target gaze tracking, pupil dilation, electromyography, neural signals, and other biometrics as emerging modalities that seeks to interface human with machine. Neural signals have varied success in robotic control for tasks due to the signal noise and artifacts that are embedded in trial data. Additionally, this raises risks of robotic damage when interfacing with costly robotic hardware due to the volatility and inconsistency that may come with neural signals. To address this complex problem, we propose a novel system that targets a virtual reality robot controlled by eye-gaze tracking and neural signals, to complete a pick and place task.

2. Introduction

i. Robotic Control

Assistive robotic computations systems that enable human motor tasks among the physically disabled has long been an area of research. Injuries to the spinal cord, strokes and neural-degenerative diseases can attribute to the need for such systems. These tasks can be classified into two categories, social interactive robotic systems, or physically assistive systems. [1] Social interactive robotic systems can be exemplified through robots that aid elderly people and specialize in assessing body language, speech and comprehension. For physical assistive robots, devices can be integrated as cortical limb prosthetics, exoskeletons [3] and rehabilitation interfaces.



ii. Gaze Tracking

Gaze tracking is commonly used for human machine interfaces. Recent studies present new methods for 3D gaze point estimation, and its integration with a robotic system architecture. [8] This allows real-time decoding, decision making, for the robot to be reactive to the gaze.

iii. EEG Brain Machine Interface Decoding

Developments in Brain-Machine Interfaces (BMI) have recently led to the ability to decode mental states from various technologies that measure neural signals. These modalities include studies in BMI literature that have used Electroencephalography (EEG) because of its non-invasive, high temporal resolution approach. EEG measures brain activity by utilizing electrodes placed on the scalp to measure voltage potentials resulting from neurons firing in the brain.

Additionally This coupled the sequence of gaze locations during virtual navigation which recorded and analyzed,[11] may allow an effective solution for gaze mediated interaction. Current gaze mediated solution for preventing midas touch include gaze fixation, blinking and external triggers. A

binary classification of EEG Control for selection may be a more seamless method for gaze mediation within a closed-loop system. This proves to be a difficult task and recent studies show that online non-invasive BMI classifiers vary in accuracy and latency. Accuracy and latency are common metrics to assess online decoders as listed in the figure below.[12]

Table 1. (Continued.)

Study	Modality	Classes	Accuracy (%)	Bits per symbol	Symbols per minute	Bits per minute (ITR)	Analysis	Population studied	Number of subjects	Comments
Müller-Putz *et al* (2006)	Tactile	2	68.0*	0.09	18.75	1.79	Online	Healthy	4	*Obtained averaging the accuracy achieved by each subject in the best session
Zhang *et al* (2007)	Tactile	2	63.00*	0.05	12	0.6	Offline	Healthy	8	*SSSEP
		2	83.00**	0.34		4.11			8	**SSSEP+SSVEP+ mu-band power
		3	61.00***	0.23		2.76			8	***SSSEP+SSVEP+ mu-band power
Brouwer and van Erp (2010)	Tactile	6	*		2.72	3.71**	Online	Healthy	11	*Performances are not clearly reported **Best condition; **the ITR was computed for each subject separately before averaging**
Kelly *et al* (2005)	Visual	2	70.3	0.12	7.5	0.91	Offline	Healthy	10	
Zhang *et al* (2010)	Visual	2	72.6	0.14	15	2.17	Online	Healthy	18	
Treder and Blankertz (2010)	Visual	36	60	2.15	2	4.29	Offline	Healthy	13	
Treder *et al* (2011)	Visual	36	91.3*	4.29	2	8.59	Online	Healthy	13	*Hex-o-Spell condition
			88.2**	4.04	2	8.08				**Cake Speller condition
			97.1***	4.83	2	9.66				***Center Speller condition
Liu *et al* (2010)	Visual	36	94.4	4.57*	1.25	5.71	Online	Healthy	8	*Random condition
			96.3	4.72**	1.25	6.02				**Searching condition
Aloise *et al* (2011)	Visual	36	77.82	3.34	2.5	8.35*	Online	Healthy	10	***The ITR was computed for each subject separately before averaging**
Acqualagna *et al* (2010)	Visual	30	90	3.9*	1.23	4.8	Offline	Healthy	9	*Best condition (SOA = 133 ms; 10 sequences)
Acqualagna and Blankertz (2011)	Visual	30	94.0*	4.29	1.2	5.14	Online	Healthy	12	*Condition no-color with SOA = 116 ms
			94.7**	4.35	1.2	5.22				**Condition color with SOA = 116 ms
			93.6***	4.25	2.4	10.20				***Condition color with SOA = 83 ms

3. Project

i. Project Design

Our target system will seek to provide a low-cost virtual robot that enables movement for impaired and healthy people to interact with a virtual environment in Unity. The virtual robot can later be reconfigured to operate or interact with physical robots using ROS, URDFs and migrating custom code to a physical experiment. We aim to develop a human-computer interactive system that merges eye movement tracking with an EEG “mouse click” to assist the user to engage with objects of interest in the virtual environment. The coupling of EEG & gaze-tracking seeks to overcome the Midas touch problem. The experiment setup will consist of an emulated virtual UR10 (Universal Robot 10) in Unity, and HTC Vive Pro for gaze tracking, an Emotiv EPOC for capturing brain frequencies and a

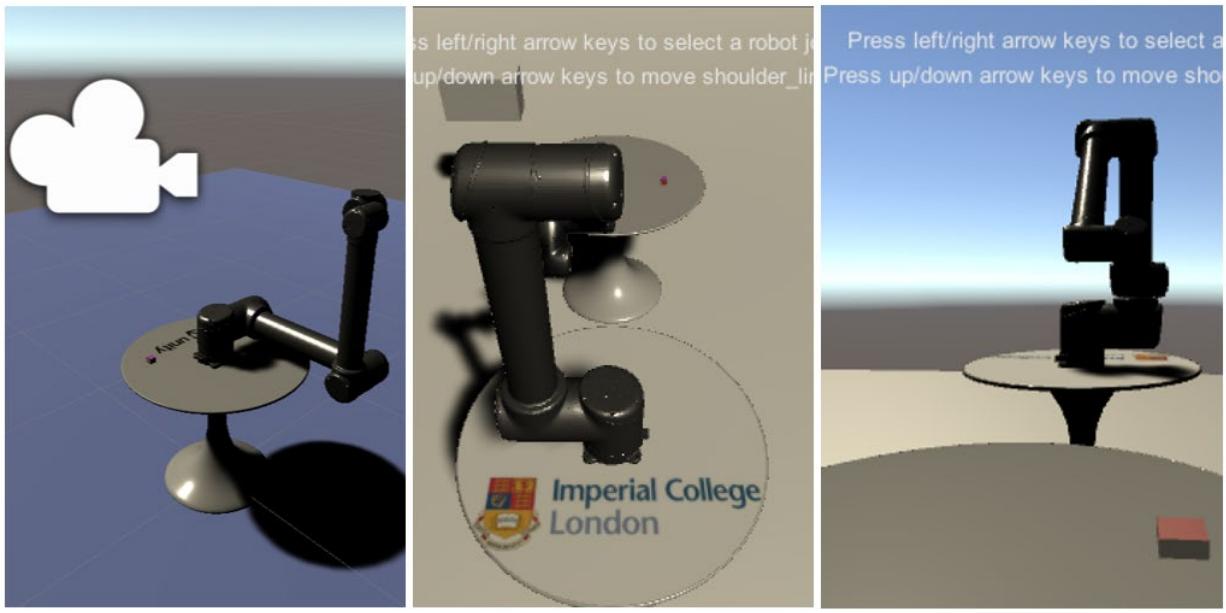


Figure 1: Left – Is a rendering of the UR10 in a Unity environment. Middle – Is the camera view of the UR10 from the observer's perspective in the virtual environment. Right – Is the subjects camera perspective which captures the objects to be selected for a pick and place experiment.

Linux/Windows machine for computational analysis. This system would have immediate impact for both patient and consumer technology.

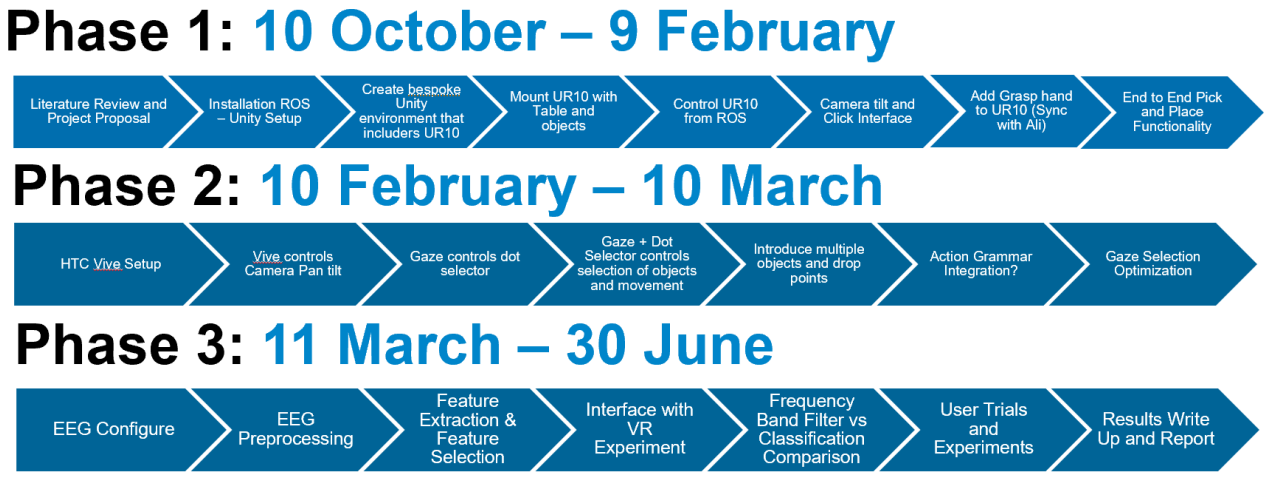
To this end we want to build a prototype system, where we create a virtual version of our gaze-based assistive robotic system in Unity for efficient experimentation on control interfaces and mechanisms. The initial robotic architecture of the system will be based on the Enhanced Robotic context aware system [8] To better fit our virtual need, multiple components will be adjusted for Unity. The Bioserve gripper hand that held in place the subject's wrist and guided it towards the target will be replaced by a simple robot crane gripper that engages a grip upon activation. Our eye-tracking equipped VR system will run experiments with users controlling the robotic system to perform object manipulation tasks for them. We will investigate the use of Eye gaze as a trigger to identify and object of interest then EEG signals as a virtual object engaging mechanism: For example, the user's gaze is synonymous to a mouse cursor to align with a desired object. The mouse "click" will be invoked using the subjects brain signals, to activate the robotic system and manipulate the desired object.

Once the full system is in place and initial testing of gaze-only control is completed, we will run data collection experiments with EEG sensors worn by the participants, to identify appropriate signals to be used as triggers: The experiment will require us to have a controlled variable to compare the system to a more practical implementation. The user will first seek to complete the task using the HTC vive head set and eye gaze for triggering the system to identify a desired object of interest. The selection of an object will be completed simply through a button (mouse, joystick, USB button) to press to execute actions. Simultaneously we will record EEG activity and then decode EEG data in an event related way to train either a machine learning decoder or a DSP filter to select the appropriate binary classifier (selected / not selected). We will adjust the experiment control with removing the preliminary trials of task completion with gaze tracking and mouse clicking with gaze tracking and

imaginary button selection. Alternatively, we will check whether other motion artifact removal can provide an improved performance then the motor. This will allow us to overcome the Midas Touch problem and effectively control real-world or VR/AR-world device by looking at it and "thinking 'click'". This gives us a 3D cursor defining methods for robotic manipulation and navigation, as well as wearable robots, self-driving vehicles etc following [2].

ii. Project Timeline

A timeline of the project experiment has been provided below. We expect to complete literature review, system configuration and the virtual environment setup on the experiment machines by February. Phase 2 is dedicated to embedding the gaze tracking into the experiment through the HTC Vive. Finally, Phase 3 is dedicated solely to building interoperability with the EEG headset, decoding and processing the signals and conducting the experiments with subjects.



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